Section Overview

- Throughput, Arithmetic Intensity, and the Roofline Model

Driving Question

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Peak throughput	35.6	$\mathrm{TFLOP/s}$
DRAM bandwidth	935.8	$\mathrm{GB/s}$

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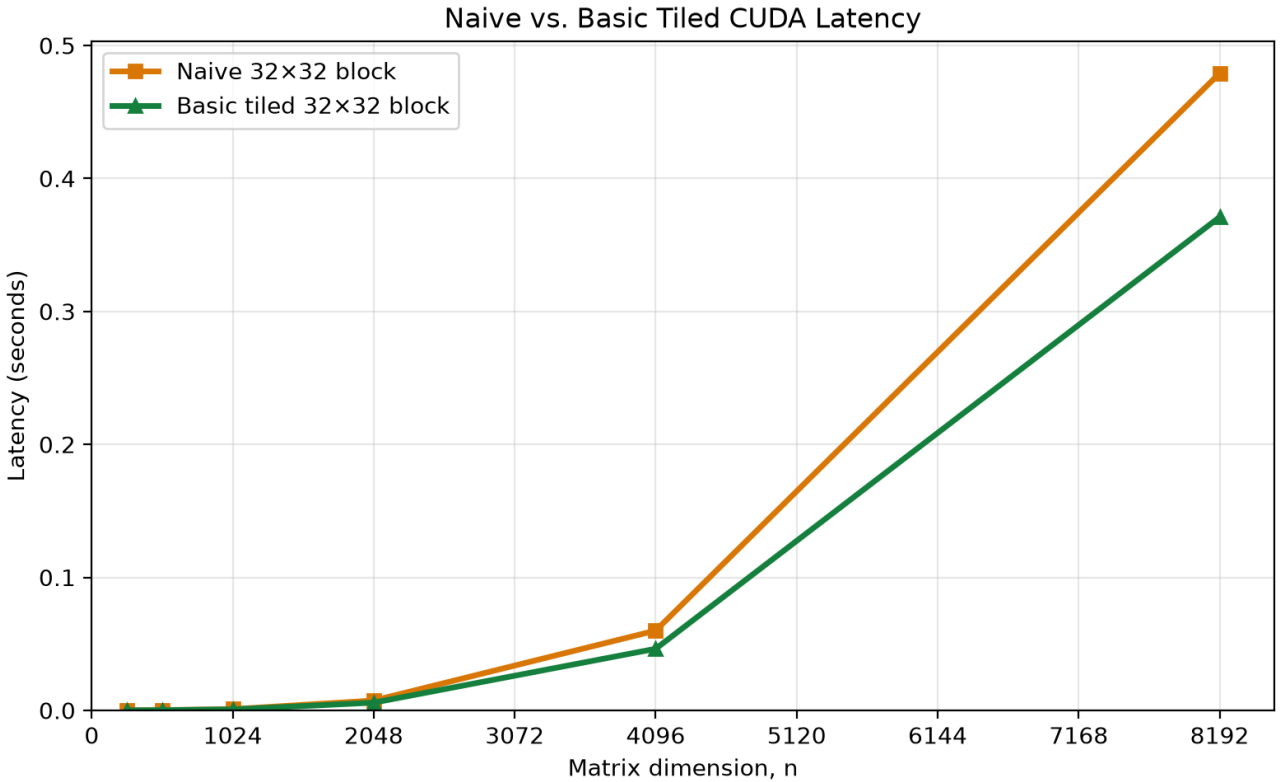
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How fast *can* we be?

Given our hardware constraints, how fast (theoretically) can we do our SGEMM operations?

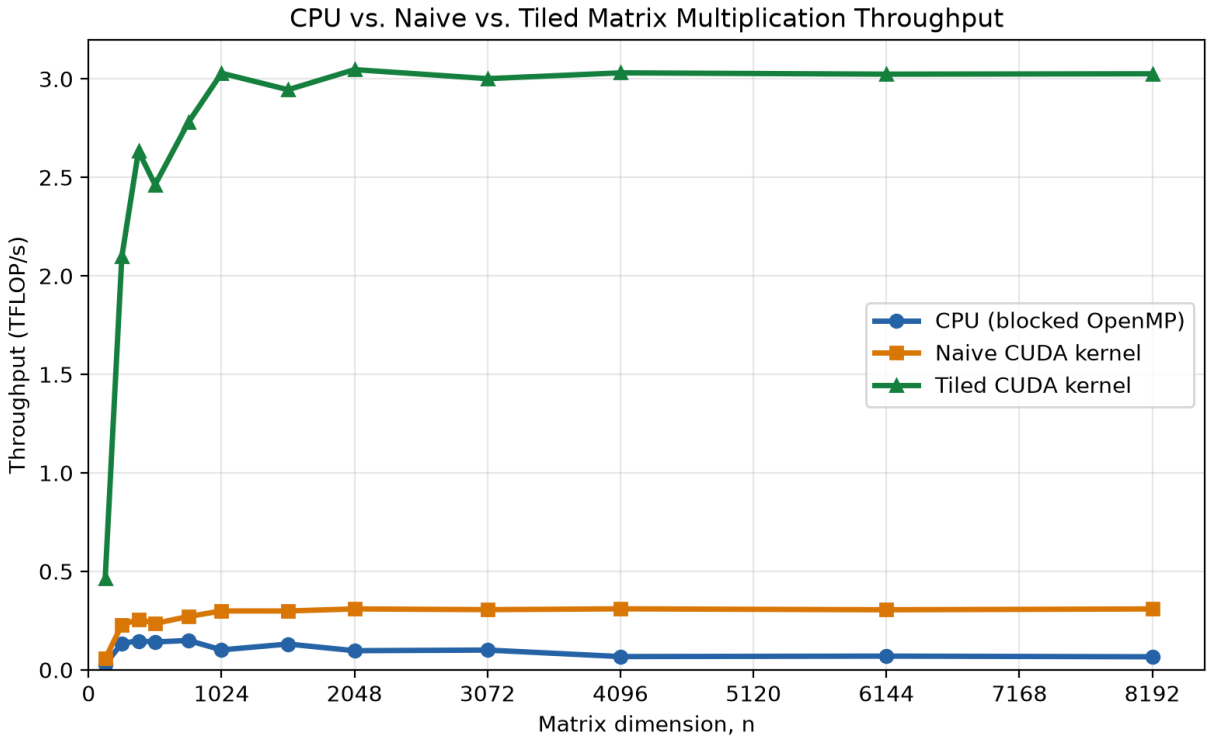
Why latency alone isn't enough

- Latency tells us how long one kernel launch takes, but it depends on the problem size.



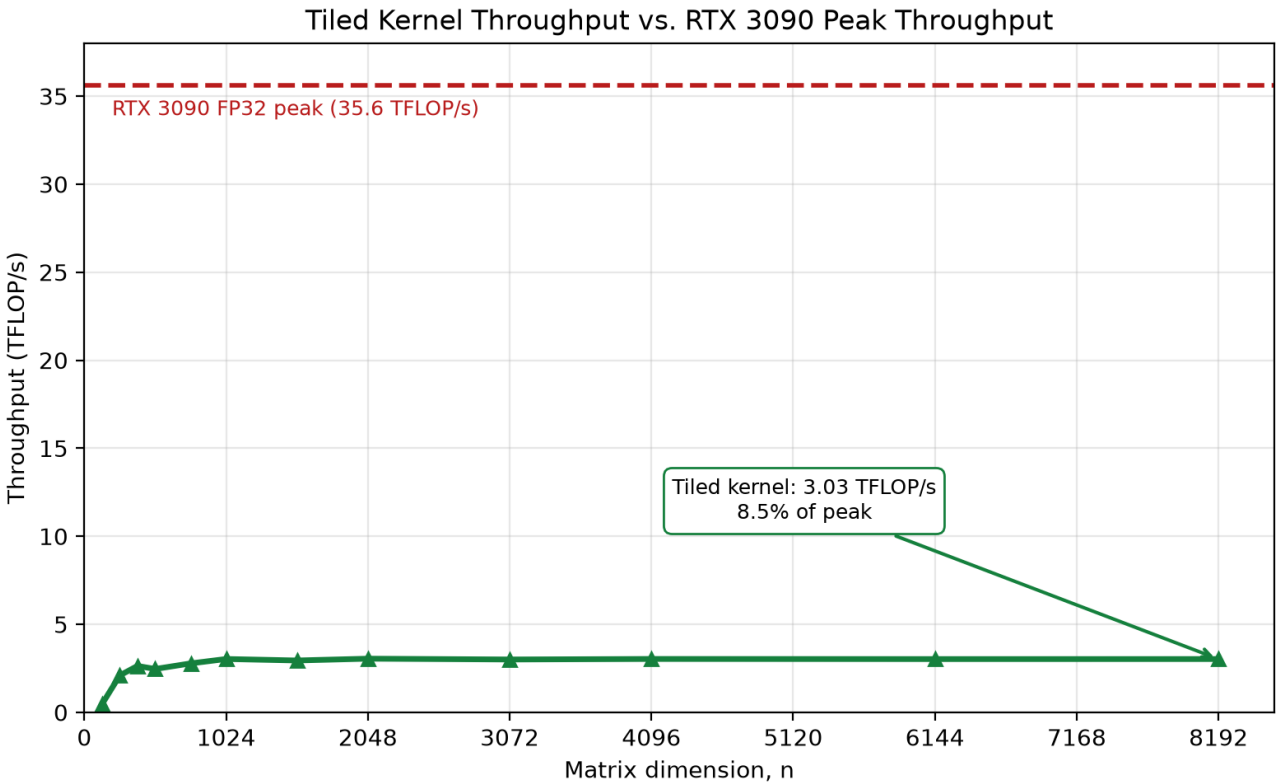
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- Since hardware is characterized by its **peak throughput**, the two values provide a first gauge of how much compute capability our kernel realizes.

Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

Throughput as a performance measure

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Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\,\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput.

Characterizing our kernels data use

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1. Throughput characterizes how well our kernel utilizes the **compute hardware**.
2. One factor that impacts final throughput is how efficiently our kernels use data.

How can we quantify how "well" our kernel uses data?

We observed that we can improve the performance of our SGEMM kernel with tiling, an optimization that improves *data use*.

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Arithmetic Intensity: Characterizing our kernels

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We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

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Example (Naive kernel)

For $n, m, k = 8192$, in the worst case, the naive kernel loads (from DRAM)

$$n \times m \times 2k \times 4 \text{ bytes} \approx 4398 \text{ GB}.$$

We do 1.10 TFLOPs of work, so we have an AI of

$$\text{AI} = \frac{1.10 \text{ TFLOP}}{4.398 \text{ TB}} \approx 0.25 \text{ FLOP/byte}.$$

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Example (Tiled kernel)

For $n, k, m = 8192$, in the worst case, our tiled kernel loads (from DRAM)

$$\frac{m}{32} \times \frac{n}{32} \times 2\left(\frac{k}{32}\right) \times 32 \times 32 \times 4 \text{ bytes} = 137.4 \text{ GB}$$

With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \text{ TFLOP}}{137.7 \text{ GB}} \approx 8.0 \text{ FLOP/byte}.$$

The Memory Roof

Memory Roof

Given the DRAM bandwidth and a kernel's AI, we say the **memory roof** of the kernel is

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In order for the CUDA Cores to do work, we have to first move the data from DRAM to the SM's.

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In order for the CUDA Cores to do work, we have to first move the data from DRAM to the SM's. If the kernel only does a small amount of work on the data (low AI), it will likely finish before the next batch of data arrives.

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Example (Intuition)

In order for the CUDA Cores to do work, we have to first move the data from DRAM to the SM's. If the kernel only does a small amount of work on the data (low AI), it will likely finish before the next batch of data arrives. **In this case, the final measured throughput will be constrained by the DRAM bandwidth.** We are **memory bound.**

The Memory Roof

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Example

Our tiling kernel has AI of 8.0 FLOP/byte, meaning it does 8 floating point operations for every byte of data moved from DRAM.
At $935.8\,\mathrm{GB/s}$ of bandwidth, our **memory roof** is

$$935.8\,\mathrm{GB/s} \times 8.0\ \mathrm{FLOP/byte} \approx 7.49\,\mathrm{TFLOP/s}.$$

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$$935.8\,\mathrm{GB/s} \times 8.0\ \mathrm{FLOP/byte} \approx 7.49\,\mathrm{TFLOP/s}.$$

This is much lower than the peak throughput of $35.6\,\mathrm{TFLOP/s}$. **No matter how we optimize this kernel, without increasing AI, we will never achieve greater than** $7.49\,\mathrm{TFLOP/s}$.

Hypothetical Kernel

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Imagine we have some kernel that has double the AI at 16.0 FLOP/s.

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$$935.8\,\mathrm{GB/s} \times 16.0\ \mathrm{FLOP/byte} \approx 15.0\,\mathrm{TFLOP/s}.$$

Since $15.0\,\mathrm{TFLOP/s} < 35.6\,\mathrm{TFLOP/s}$, this kernel will still be memory bound.

Hypothetical Kernels

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AI (FLOP/byte)	Memory Roof (TFLOP/s)
8	7.49
16	14.97
32	29.95
36	33.69
38	35.56
40	37.43

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38	35.56
40	37.43

Wait! Our peak throughput is 35.6 $\mathrm{TFLOP/s}$! At AI ≥ 38, we are **compute bound**. Our data can finally keep up with the CUDA Cores!

Ridge Point

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AI $= 38$ FLOP/byte is known as the **ridge point** *RP*, and can be found by solving:

$$\text{Data roof} = \text{Peak throughput}$$

$$\text{DRAM Bandwidth} \times \text{RP} = \text{Peak throughput}$$

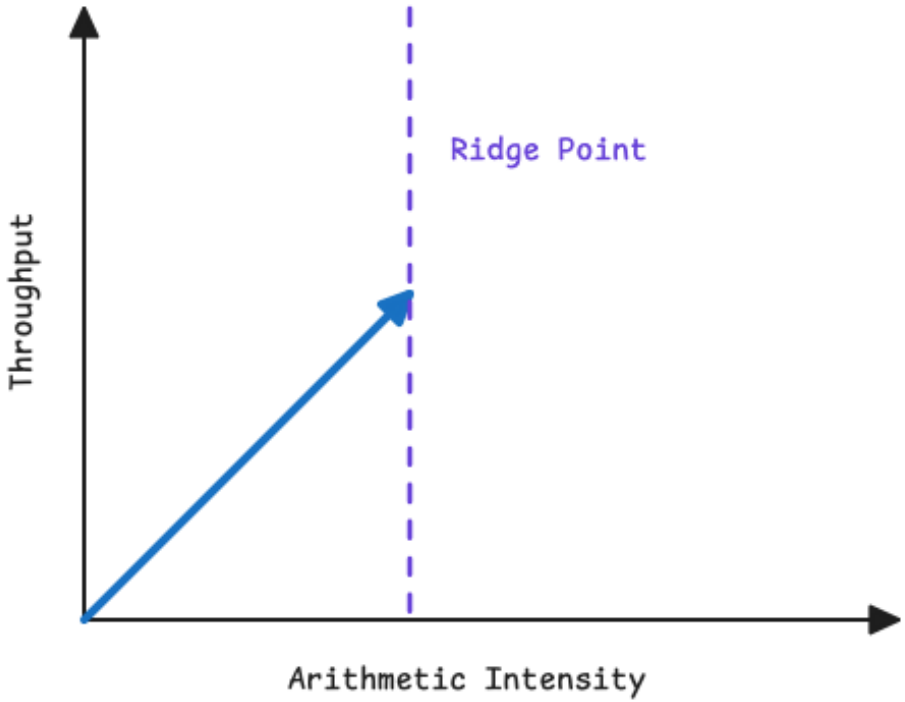
$$\text{RP} = \frac{\text{Peak throughput}}{\text{DRAM Bandwidth}}$$

$$\text{RP} = \frac{35.6\,\mathrm{TFLOP/s}}{935.8\,\mathrm{GB/s}} \approx 38.0 \text{ FLOP/s}$$

Roofline Model

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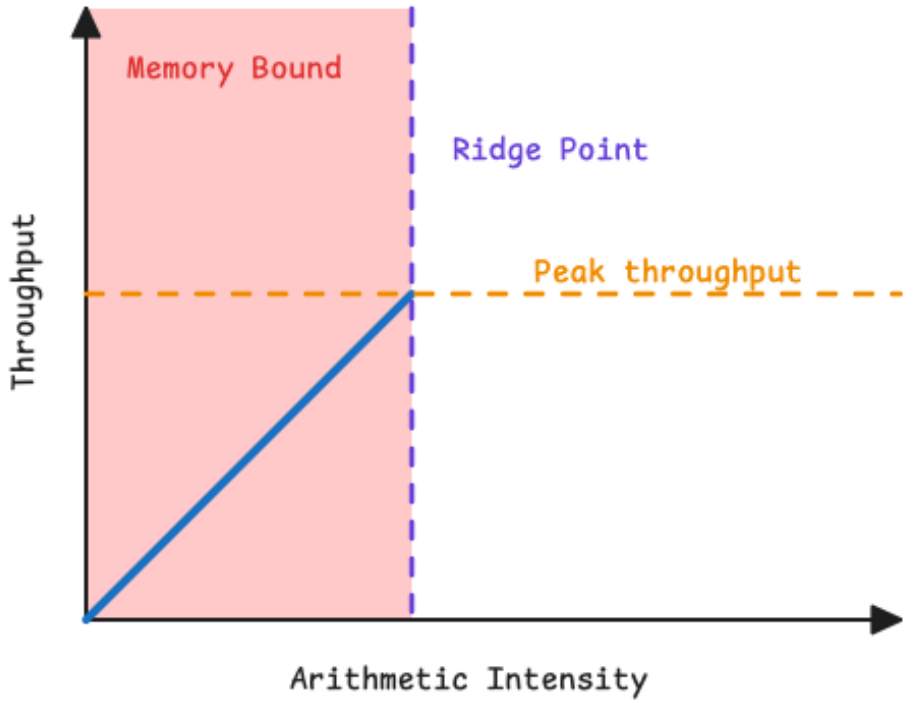
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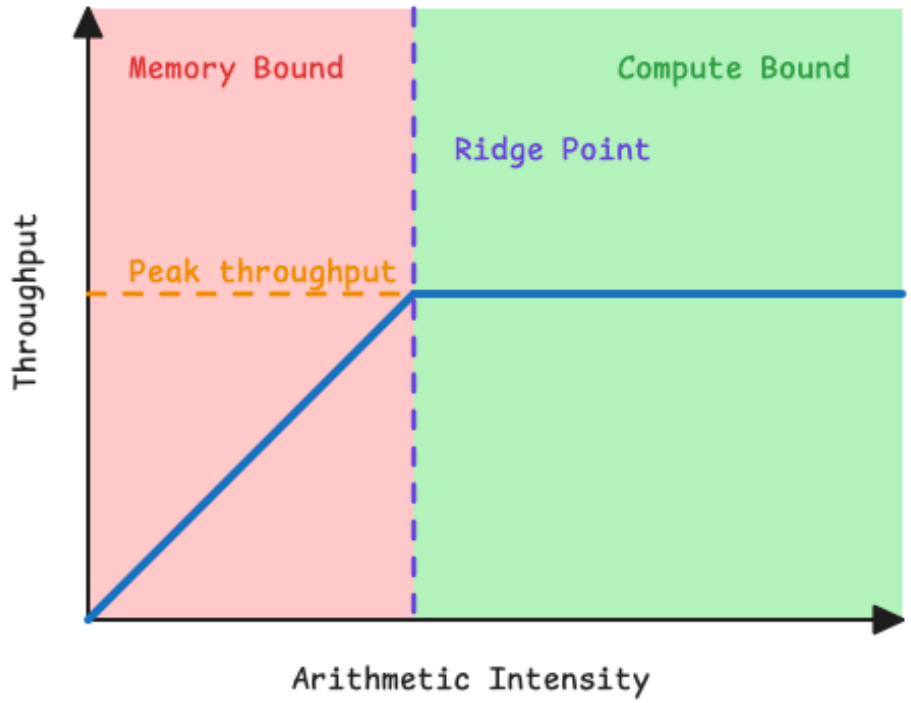
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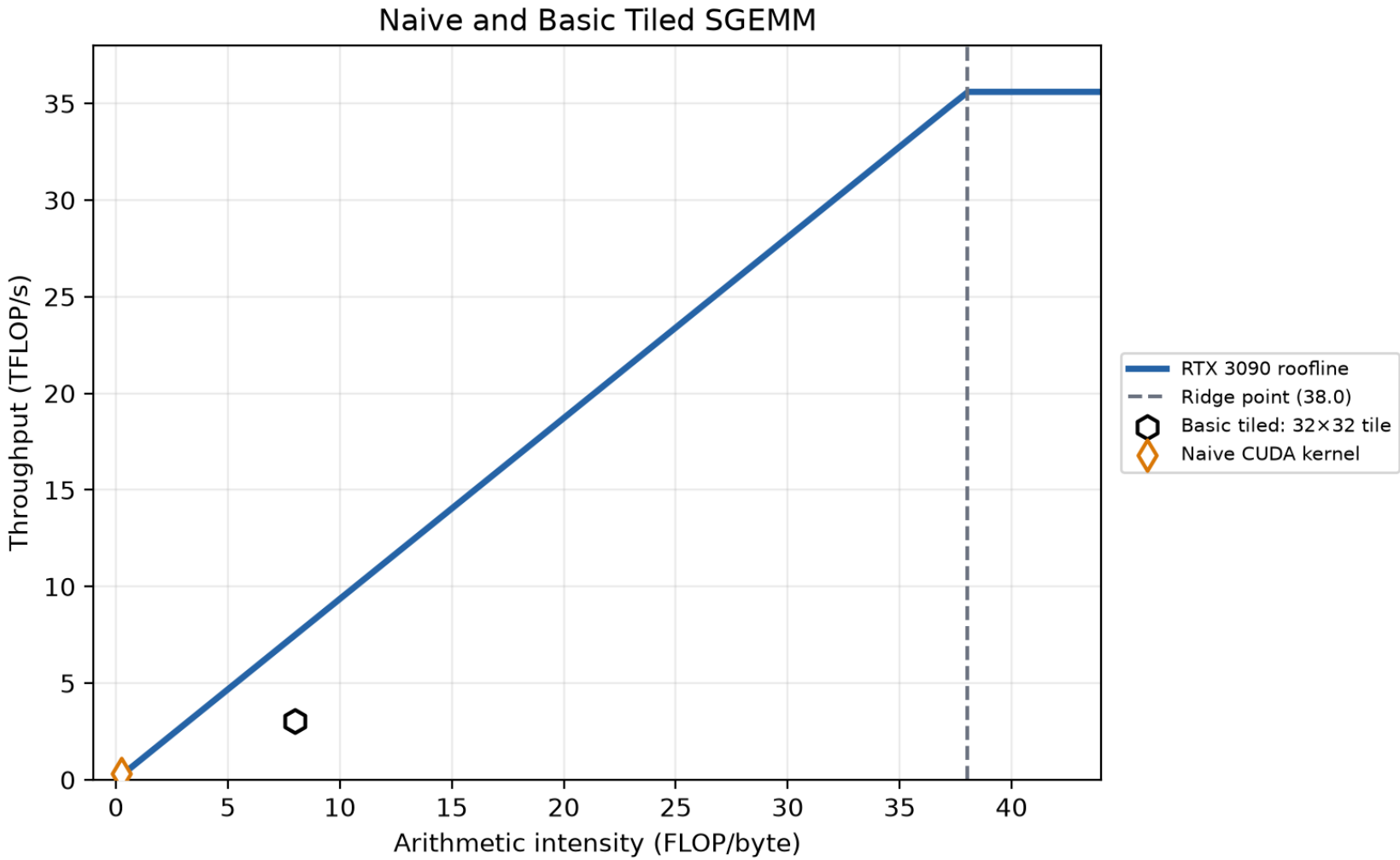
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Roofline Model of our RTX 3090



Since we are memory bound, increasing AI should increase performance.